

Topics: 16.7 Surface Integrals; 16.9 The Divergence Theorem

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1. Evaluate the following scalar surface integrals.

- (a) $\iint_S x^2 y z \, dS$, where S is the part of the plane $z = 1 + 2x + 3y$ that lies above $[0, 3] \times [0, 2]$.

The surface is given explicitly by

$$z = g(x, y) = 1 + 2x + 3y,$$

as (x, y) ranges over $D = [0, 3] \times [0, 2]$. Taking partial derivatives, $g_x = 2$ and $g_y = 3$. Thus

$$dS = \sqrt{1 + g_x^2 + g_y^2} \, dA = \sqrt{1 + 2^2 + 3^2} \, dA = \sqrt{14} \, dA.$$

Therefore,

$$\begin{aligned} \iint_S x^2 y z \, dS &= \int_0^3 \int_0^2 x^2 y (1 + 2x + 3y) \sqrt{14} \, dy \, dx \\ &= \sqrt{14} \int_0^3 x^2 \int_0^2 y (1 + 2x + 3y) \, dy \, dx \\ &= \sqrt{14} \int_0^3 x^2 (10 + 4x) \, dx \\ &= \sqrt{14} \int_0^3 (10x^2 + 4x^3) \, dx \\ &= \sqrt{14} \left[\frac{10}{3} x^3 + x^4 \right]_0^3 \\ &= 171\sqrt{14}. \end{aligned}$$

- (b) $\iint_S y^2 z^2 \, dS$, where S is the part of the cone $y = \sqrt{x^2 + z^2}$ with $0 \leq y \leq 5$.

We parametrize the cone by

$$\mathbf{r}(r, \theta) = \langle r \cos \theta, r, r \sin \theta \rangle, \quad 0 \leq r \leq 5, \quad 0 \leq \theta \leq 2\pi.$$

Taking partial derivatives,

$$\mathbf{r}_r = \langle \cos \theta, 1, \sin \theta \rangle \quad \text{and} \quad \mathbf{r}_\theta = \langle -r \sin \theta, 0, r \cos \theta \rangle.$$

Taking the cross product, one finds that

$$\mathbf{r}_r \times \mathbf{r}_\theta = \langle r \cos \theta, -r, r \sin \theta \rangle,$$

so

$$|\mathbf{r}_r \times \mathbf{r}_\theta| = \sqrt{r^2 \cos^2 \theta + r^2 + r^2 \sin^2 \theta} = \sqrt{2} r.$$

On the surface, $y = r$ and $z = r \sin \theta$. Therefore,

$$\begin{aligned}\iint_S y^2 z^2 dS &= \int_0^{2\pi} \int_0^5 r^2 (r \sin \theta)^2 \sqrt{2} r dr d\theta \\ &= \sqrt{2} \int_0^{2\pi} \int_0^5 r^5 \sin^2 \theta dr d\theta \\ &= \sqrt{2} \left(\int_0^5 r^5 dr \right) \left(\int_0^{2\pi} \sin^2 \theta d\theta \right) \\ &= \sqrt{2} \left[\frac{r^6}{6} \right]_0^5 (\pi) \\ &= \frac{15625\pi\sqrt{2}}{6}.\end{aligned}$$

- (c) $\iint_S z dS$, where S is given by $\mathbf{r}(u, v) = u^2 \mathbf{i} + v \mathbf{j} + u \mathbf{k}$ as (u, v) ranges over $[0, 1] \times [0, 1]$.

The surface is given by

$$\mathbf{r}(u, v) = \langle u^2, v, u \rangle,$$

as (u, v) ranges over $D = [0, 1] \times [0, 1]$. Taking partial derivatives,

$$\mathbf{r}_u = \langle 2u, 0, 1 \rangle \quad \text{and} \quad \mathbf{r}_v = \langle 0, 1, 0 \rangle.$$

Taking the cross product, one finds that

$$\mathbf{r}_u \times \mathbf{r}_v = \langle -1, 0, 2u \rangle,$$

and so

$$|\mathbf{r}_u \times \mathbf{r}_v| = \sqrt{1 + 4u^2}.$$

Since $z = u$, we get

$$\iint_S z dS = \int_0^1 \int_0^1 u \sqrt{1 + 4u^2} dv du = \int_0^1 u \sqrt{1 + 4u^2} du.$$

Let $w = 1 + 4u^2$, so $dw = 8u du$. When $u = 0$, $w = 1$, and when $u = 1$, $w = 5$. Therefore,

$$\iint_S z dS = \frac{1}{8} \int_1^5 w^{1/2} dw = \frac{1}{8} \left[\frac{2}{3} w^{3/2} \right]_1^5 = \frac{5\sqrt{5} - 1}{12}.$$

2. Evaluate the flux integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ for the given vector field $\mathbf{F}$ and oriented surface S .

- (a) Let $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$ and let S be the surface parametrized by $\mathbf{r}(u, v) = \langle u, v, u^2 + v^2 \rangle$, for $0 \leq u \leq 1$ and $0 \leq v \leq 1$, with upward orientation.

We are given $\mathbf{r}(u, v) = \langle u, v, u^2 + v^2 \rangle$. Taking partial derivatives,

$$\mathbf{r}_u = \langle 1, 0, 2u \rangle \quad \text{and} \quad \mathbf{r}_v = \langle 0, 1, 2v \rangle.$$

Taking the cross product, one finds that

$$\mathbf{r}_u \times \mathbf{r}_v = \langle -2u, -2v, 1 \rangle.$$

Since this vector has positive z -component, it gives the upward orientation. Therefore,

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \int_0^1 \int_0^1 \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, dv \, du \\ &= \int_0^1 \int_0^1 \langle u, v, u^2 + v^2 \rangle \cdot \langle -2u, -2v, 1 \rangle \, dv \, du \\ &= \int_0^1 \int_0^1 (-2u^2 - 2v^2 + u^2 + v^2) \, dv \, du \\ &= \int_0^1 \int_0^1 (-u^2 - v^2) \, dv \, du \\ &= \int_0^1 \left[-u^2 v - \frac{v^3}{3} \right]_{v=0}^{v=1} du \\ &= \int_0^1 \left(-u^2 - \frac{1}{3} \right) du \\ &= \left[-\frac{u^3}{3} - \frac{u}{3} \right]_{u=0}^{u=1} \\ &= -\frac{1}{3} - \frac{1}{3} \\ &= -\frac{2}{3}. \end{aligned}$$

- (b) Let $\mathbf{F}(x, y, z) = \langle y, -x, 2z \rangle$ and let S be the spherical cap $x^2 + y^2 + z^2 = 4$ with $z \geq 1$, with downward orientation.

We write the surface as

$$z = g(x, y) = \sqrt{4 - x^2 - y^2}.$$

Since S is the spherical cap with $z \geq 1$, its projection onto the xy -plane satisfies

$$\sqrt{4 - x^2 - y^2} \geq 1.$$

Thus,

$$4 - x^2 - y^2 \geq 1,$$

so

$$D = \{(x, y) \mid x^2 + y^2 \leq 3\}.$$

For downward orientation, we use

$$d\mathbf{S} = \langle g_x, g_y, -1 \rangle dA.$$

Taking partial derivatives,

$$g_x = -\frac{x}{\sqrt{4 - x^2 - y^2}} \quad \text{and} \quad g_y = -\frac{y}{\sqrt{4 - x^2 - y^2}}.$$

Thus,

$$\begin{aligned}
 \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_D (Pg_x + Qg_y - R) dA \\
 &= \iint_D \left[y \left(-\frac{x}{\sqrt{4-x^2-y^2}} \right) + (-x) \left(-\frac{y}{\sqrt{4-x^2-y^2}} \right) - 2\sqrt{4-x^2-y^2} \right] dA \\
 &= \iint_D -2\sqrt{4-x^2-y^2} dA \\
 &= -2 \int_0^{2\pi} \int_0^{\sqrt{3}} \sqrt{4-r^2} r dr d\theta \\
 &= -2 \left(\int_0^{2\pi} d\theta \right) \left(\int_0^{\sqrt{3}} \sqrt{4-r^2} r dr \right) \\
 &= -4\pi \int_0^{\sqrt{3}} r \sqrt{4-r^2} dr \\
 &= -4\pi \left[-\frac{1}{3}(4-r^2)^{3/2} \right]_0^{\sqrt{3}} \\
 &= -4\pi \left(-\frac{1}{3} + \frac{8}{3} \right) \\
 &= -4\pi \left(\frac{7}{3} \right) \\
 &= -\frac{28\pi}{3}.
 \end{aligned}$$

3. Use the Divergence Theorem to find the outward flux of $\mathbf{F}$ across the closed surface S .

- (a) Let $\mathbf{F}(x, y, z) = \langle xye^z, xy^2z^3, -ye^z \rangle$ and let S be the boundary surface of the rectangular box bounded by the coordinate planes and the planes $x = 3$, $y = 2$, and $z = 1$.

We compute

$$\begin{aligned}
 \operatorname{div} \mathbf{F} &= \frac{\partial}{\partial x}(xye^z) + \frac{\partial}{\partial y}(xy^2z^3) + \frac{\partial}{\partial z}(-ye^z) \\
 &= ye^z + 2xyz^3 - ye^z \\
 &= 2xyz^3.
 \end{aligned}$$

Therefore, by the Divergence Theorem,

$$\begin{aligned}
 \iint_S \mathbf{F} \cdot d\mathbf{S} &= \int_0^3 \int_0^2 \int_0^1 2xyz^3 dz dy dx \\
 &= 2 \left(\int_0^3 x dx \right) \left(\int_0^2 y dy \right) \left(\int_0^1 z^3 dz \right) \\
 &= 2 \left(\frac{9}{2} \right) (2) \left(\frac{1}{4} \right) \\
 &= \frac{9}{2}.
 \end{aligned}$$

- (b) Let $\mathbf{F}(x, y, z) = \langle xe^y, z - e^y, -xy \rangle$. Let S be the ellipsoid $x^2 + 2y^2 + 3z^2 = 4$.

We compute

$$\begin{aligned}\operatorname{div} \mathbf{F} &= \frac{\partial}{\partial x}(xe^y) + \frac{\partial}{\partial y}(z - e^y) + \frac{\partial}{\partial z}(-xy) \\ &= e^y - e^y + 0 \\ &= 0.\end{aligned}$$

Therefore, by the Divergence Theorem,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E 0 \, dV = 0.$$

- (c) Let $\mathbf{F}(x, y, z) = z\mathbf{i} + y\mathbf{j} + x\mathbf{k}$ and S be the boundary surface of the upper half-ball $x^2 + y^2 + z^2 \leq 1$, with $z \geq 0$, including the disk in the plane $z = 0$.

We compute

$$\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(z) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(x) = 0 + 1 + 0 = 1.$$

Therefore, by the Divergence Theorem,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E 1 \, dV = \operatorname{Vol}(E) = \frac{1}{2} \left(\frac{4}{3} \pi (1)^3 \right) = \frac{2\pi}{3}.$$

- (d) Let $\mathbf{F}(x, y, z) = xy\mathbf{i} + y^2\mathbf{j} + z\mathbf{k}$ and let S be the boundary surface of the tetrahedron bounded by the plane $x + 2y + z = 2$, and the three coordinate planes.

We compute

$$\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(y^2) + \frac{\partial}{\partial z}(z) = y + 2y + 1 = 3y + 1.$$

The tetrahedron is given by

$$x + 2y + z \leq 2, \quad x, y, z \geq 0.$$

When integrating, we will use y as the outer variable. For a fixed y , we have

$$0 \leq x \leq 2 - 2y, \quad 0 \leq z \leq 2 - 2y - x.$$

Therefore, by the Divergence Theorem,

$$\begin{aligned}
 \iint_S \mathbf{F} \cdot d\mathbf{S} &= \int_0^1 \int_0^{2-2y} \int_0^{2-2y-x} (3y+1) dz dx dy \\
 &= \int_0^1 (3y+1) \int_0^{2-2y} \int_0^{2-2y-x} dz dx dy \\
 &= \int_0^1 (3y+1) \int_0^{2-2y} (2-2y-x) dx dy \\
 &= \int_0^1 (3y+1) \left[(2-2y)x - \frac{x^2}{2} \right]_0^{2-2y} dy \\
 &= \int_0^1 (3y+1) \cdot \frac{(2-2y)^2}{2} dy \\
 &= \frac{1}{2} \int_0^1 (3y+1)(2-2y)^2 dy \\
 &= \frac{1}{2} \int_0^1 (12y^3 - 20y^2 + 4y + 4) dy \\
 &= \frac{1}{2} \left[3y^4 - \frac{20}{3}y^3 + 2y^2 + 4y \right]_0^1 \\
 &= \frac{7}{6}.
 \end{aligned}$$

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Surface Integrals of Scalar Functions. Suppose f is a function of three variables whose domain includes a smooth surface S . If S is given parametrically by

$$\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k},$$

and S is covered only once as (u, v) ranges over D , then

$$\iint_S f(x, y, z) dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| dA.$$

If S is given explicitly by $z = g(x, y)$ for $(x, y) \in D$, then

$$\iint_S f(x, y, z) dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} dA.$$

Surface Integrals of Vector Fields. Let S be an oriented surface with unit normal vector $\mathbf{n}$. The flux of a vector field $\mathbf{F}$ across S is

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS.$$

If S is parametrized by

$$\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}, \quad (u, v) \in D,$$

and is covered only once, then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, dA,$$

using the orientation determined by $\mathbf{r}_u \times \mathbf{r}_v$. For the opposite orientation, use $-(\mathbf{r}_u \times \mathbf{r}_v)$. If S is the graph $z = g(x, y)$ with $(x, y) \in D$, then for upward orientation,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F}(x, y, g(x, y)) \cdot \langle -g_x, -g_y, 1 \rangle \, dA.$$

For downward orientation, instead use $\langle g_x, g_y, -1 \rangle$.

The Divergence Theorem. Let E be a simple solid region, or a finite union of simple solid regions, and let S be the boundary surface of E , given with the positive (outward) orientation. Let $\mathbf{F}$ be a vector field whose component functions have continuous first-order partial derivatives on an open region containing E . Then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E \operatorname{div} \mathbf{F} \, dV.$$

Suggested Textbook Problems

Section 16.7: 1-32

Section 16.9: 1-17, 19-24